$$M x_{min}^2 = \int_0^\theta x^2 \cdot n\left(1-\frac{x}{\theta}\right)^{n-1}\cdot\frac{1}{\theta}\,dx = \left\{\begin{matrix} t=1-\frac{x}{\theta} \\ dx=-\theta\,dt \end{matrix};\ x=(1-t)\theta\right\} =$$

$$= \int_1^0 -(1-t)^2\theta^2\cdot n\frac{(t)^{n-1}}{\theta}\theta\,dt = \int_0^1 (1-t)^2\theta^2 n t^{n-1}\,dt = \theta^2 n\left(\int_0^1 (t^{n-1}-2t^n+t^{n+1})\,dt\right)$$

$$= \theta^2 n\left(\frac{t^n}{n}\Big|_0^1 - \frac{2t^{n+1}}{n+1}\Big|_0^1 + \frac{t^{n+2}}{n+2}\Big|_0^1\right) = \theta^2 n\left(\frac{1}{n}-\frac{2}{n+1}+\frac{1}{n+2}\right) =$$

$$= \frac{\theta^2 n\left(\cancel{n}(n+1)(n+2)-2n(n+2)+n(n+1)\right)}{n(n+1)(n+2)} = \frac{2\theta^2\cancel{n}}{\cancel{n}(n+1)(n+2)} = \frac{2\theta^2}{(n+1)(n+2)};$$

$$\cancel{n^2}+\cancel{3n}+2-\cancel{2n^2}-\cancel{4n}+\cancel{n^2}+\cancel{n}=2$$

Тогда $D\tilde{\theta}_2' = (n+1)^2\left(\frac{2\theta^2}{(n+1)(n+2)}-\frac{\theta^2}{(n+1)^2}\right) = \frac{2\theta^2(n+1)}{n+2}-\theta^2 =$

$$= \theta^2\left(\frac{2(n+1)}{n+2}-1\right) = \theta^2\left(\frac{2n+2-n-2}{n+2}\right) = \theta^2\left(\frac{n}{n+2}\right) \not\xrightarrow[n\to\infty]{} 0$$

Достаточное условие в данной ситуации не выполняется.

Тогда найдём через определение:

$\forall\theta>0\ \forall\varepsilon>0\quad P(|\tilde{\theta}-\theta|\geq\varepsilon)\xrightarrow[n\to\infty]{}0$; $\Phi_2(x)=1-(1-F(x))^n$

~~т.е. $\forall\theta>0$~~

1) $P(|\tilde{\theta}_2'-\theta|\geq\varepsilon) = P(\tilde{\theta}_2'\geq\theta+\varepsilon) = P((n+1)x_{min}\geq\theta+\varepsilon) =$

$$= P\left(x_{min}\geq\frac{\theta+\varepsilon}{n+1}\right) = 1-P\left(x_{min}<\frac{\theta+\varepsilon}{n+1}\right) =$$

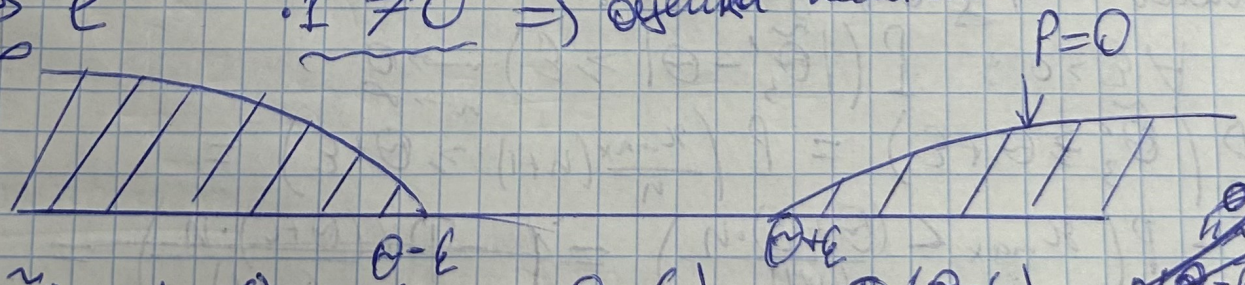
$$= 1-\Phi\left(\frac{\theta+\varepsilon}{n+1}\right)$$

$$\Phi(x) = 1-(1-F(x))^n \Rightarrow P\left(x_{min}\geq\frac{\theta+\varepsilon}{(n+1)}\right) =$$

$$= 1-(1-(1-F(x)))^n = \cancel{1-1+}\left(1-\frac{\theta+\varepsilon}{\theta(n+1)}\right)^n = \left(1-\frac{\theta+\varepsilon}{\theta(n+1)}\right)^n =$$

$$= \cancel{\left(1+\frac{-\theta+\varepsilon}{(n+1)\theta}\right)^{\frac{(n+1)\theta}{(\theta+\varepsilon)}\cdot\frac{1}{\theta}(\theta+\varepsilon)\cdot(-1)}-1} \xrightarrow[n\to\infty]{} \left(1+\frac{-(\theta+\varepsilon)}{(n+1)\theta}\right)^{\frac{-(n+1)\theta}{(\theta+\varepsilon)\theta}(\theta+\varepsilon)(-1)}-1 \xrightarrow[n\to\infty]{}$$

$$\xrightarrow[n\to\infty]{} e^{-\frac{\theta+\varepsilon}{\theta}}\cdot 1 \neq 0 \Rightarrow \text{оценка несостоятельная}$$



2) $P(|\tilde{\theta}_2^*-\theta|\geq\varepsilon) = P\left(x_{min}\leq\frac{\theta-\varepsilon}{n+1}\right) = \Phi\left(\frac{\theta-\varepsilon}{n+1}\right) + \cancel{P\left(\frac{\theta-\varepsilon}{n+1}\right)} =$

$$= 1-\left(1-F\left(\frac{\theta-\varepsilon}{n+1}\right)\right)^n = 1-\left(1-\frac{\theta-\varepsilon}{\theta}\right)^n \xrightarrow[n\to\infty]{} \left\{1-\frac{\theta-\varepsilon}{\theta}=\frac{\varepsilon}{\theta}\right\}\to 1,$$

т.к. $\exists\varepsilon>0$ и $\exists\theta>0$